out the details of such a chain reaction. Zel'dovich and Khariton—already among the world's best experts on chemical explosions—leaped on the problem. Within a few months they had shown (as, in parallel, did others in the West) that such an explosion cannot occur in nature, because naturally occurring uranium consists mostly of uranium-238 and not enough uranium-235. However, they concluded, if one were to artificially separate out uranium-235 and concentrate it, then one could make a chain-reaction explosion. (The Americans would soon embark on such separation to make the fuel for their Hiroshima bomb.) The curtain of secrecy had not yet descended around nuclear research, so Zel'dovich and Khariton published their calculations in the most prestigious of Soviet physics journals, the *Journal of Experimental and Theoretical Physics*, for all the world to see.

During the six years of World War II, physicists of the warring nations developed sonar, mine sweepers, rockets, radar, and, most fatefully, the atomic bomb. Oppenheimer led the "Manhattan Project" at Los Alamos, New Mexico, to design and build the American bombs. Wheeler was the lead scientist in the design and construction of the world's first production-scale nuclear reactors, in Hanford, Washington, which made the plutonium-239 for the Nagasaki bomb.

After the bombs' decimation of Hiroshima and Nagasaki and the deaths of several hundred thousand people, Oppenheimer was in anguish: "If atomic bombs are to be added to the arsenals of a warring world, or to the arsenals of nations preparing for war, then the time will come when mankind will curse the name of Los Alamos and Hiroshima." "In some sort of crude sense which no vulgarity, no humor, no overstatement can quite extinguish, the physicists have known sin; and this is a knowledge which they cannot lose."

But Wheeler had the opposite kind of regret: "As I look back on [1939 and my fission theory work with Bohr], I feel a great sadness. How did it come about that I looked on fission first as a physicist [simply curious to know how fission works], and only secondarily as a citizen [intent on defending my country]? Why did I not look at it first as a citizen and only secondarily as a physicist? A simple survey of the records shows that between twenty and twenty-five million people perished in World War 11 and more of them in the later years than in the earlier years. Every month by which the war was shortened would have meant a saving of the order of